Neutrino Mass Origin from Recognition Science: Deep φ-Ladder Rungs, Dirac Nature, and Normal Hierarchy

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Abstract

We derive the origin of neutrino masses within Recognition Science (RS). Neutrino masses arise from the same master mass law as all other fermions, but neutrinos occupy deep negative rungs on the φ-ladder ($R \approx -54$ to -60), explaining their sub-eV masses without a seesaw mechanism. Since neutrinos are electrically neutral ($Z = 0$), their gap function vanishes ($\mathrm{gap}(0) = 0$), simplifying the mass law. The fractional rung residues are $r_{\nu_3}^{\mathrm{frac}} = -217/4$, $r_{\nu_2}^{\mathrm{frac}} = -231/4$, and $r_{\nu_1}^{\mathrm{frac}} = -239/4$, yielding masses $m_3 \approx 0.056$ eV, $m_2 \approx 0.008$ eV, and $m_1 \approx 0.003$ eV, with generation spacing $\Delta \approx 4$ rungs. The squared mass differences are compatible with oscillation data. Neutrinos are predicted to be Dirac particles (from double-entry ledger conservation), follow normal mass ordering (from ladder monotonicity), and have a mass sum $\sum m_\nu \approx 0.067$ eV.

1 Introduction

Neutrino oscillation experiments have established that neutrinos have non-zero masses [3], but three critical properties remain unknown:

1. The **absolute mass scale** (m_{ν_1} is unconstrained by oscillations).
2. The **mass ordering**: normal ($m_1 < m_2 < m_3$) or inverted ($m_3 < m_1 < m_2$).
3. The **Dirac vs. Majorana nature**: whether $\nu = \bar{\nu}$ (Majorana) or $\nu \neq \bar{\nu}$ (Dirac).

In the Standard Model, neutrino masses require either right-handed neutrinos (Dirac mass via Yukawa coupling) or lepton number violation (Majorana mass via the seesaw mechanism [4]). Neither option is determined by the SM itself.

RS [2] provides definitive answers to all three questions from the φ-ladder structure.

2 Recognition Science Framework

RS derives all physical structure from three axioms [2]:

(A1) $J(1) = 0$ *(Normalization)*

(A2) $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$ *(Recognition Composition Law)*

(A3) $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$ *(Calibration)*

Theorem 2.1 (Cost Uniqueness [2]). *The unique continuous solution is $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, satisfying $J(x) \geq 0$ with equality iff $x = 1$.*

Proof sketch. Setting $H(t) = 1 + J(e^t)$ transforms (A2) into the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$ with $H(0) = 1$ and $H''(0) = 1$ from (A1) and (A3). By the Aczél classification [1], $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

Theorem 2.2 (φ Forcing). *The discrete ledger's self-similarity constraint forces the scaling ratio $\varphi = (1 + \sqrt{5})/2$, the unique positive root of $x^2 = x + 1$.*

Proof. The coherence energy $E_{\text{coh}} = \varphi^{-5}$ requires the scaling ratio to satisfy $\varphi^2 = \varphi + 1$. Solving: $(1 \pm \sqrt{5})/2$; only $(1 + \sqrt{5})/2 > 0$. □

All particle masses are organized on the φ-ladder: $m = Y_s \cdot E_{\text{coh}} \cdot \varphi^r$, where Y_s is the sector yardstick, $E_{\text{coh}} = \varphi^{-5}$ is the coherence quantum, and r is an integer rung determined by the $D = 3$ cube geometry, generation assignments, and charge gaps.

3 The Master Mass Law for Neutrinos

Definition 3.1 (General Mass Law). All fermion masses in RS follow the universal mass law at the anchor scale $\mu^\star = 182.2$ GeV:

$$m_f = Y_s \cdot E_{\text{coh}} \cdot \varphi^{r_f - 8 + \text{gap}(Z_f) + f_f^{\text{RG}}}, \tag{1}$$

where Y_s is the sector yardstick, $E_{\text{coh}} = \varphi^{-5}$ is the coherence quantum, $\varphi = (1 + \sqrt{5})/2$, r_f is the integer rung, $\text{gap}(Z_f) = Z_f + Z_f^2 + Z_f^4$ is the charge-dependent correction (with $Z_f = 6Q_f$), and f_f^{RG} is the renormalization group transport residue.

Corollary 3.2 (Neutrino Mass Simplification). *For neutrinos, $Q = 0$ implies $Z = 6Q = 0$, so:*

$$\text{gap}(0) = 0 + 0 + 0 = 0. \tag{2}$$

The neutrino mass law simplifies to:

$$\boxed{m_{\nu_i} = Y_\nu \cdot E_{\text{coh}} \cdot \varphi^{r_{\nu_i} - 8 + f_{\nu_i}^{\text{RG}}}.} \tag{3}$$

Remark 3.3. The vanishing of the gap function for neutrinos is the fundamental reason for their simplicity: neutrinos are the "bare" φ-ladder excitations, without the charge-dependent corrections that modify charged fermion masses. This also explains the 4-rung generation spacing (compared to the more complex spacings for charged leptons), since the spacing is determined purely by the geometric generation torsion without electromagnetic corrections.

4 Deep Negative Rung Assignments

Definition 4.1 (Neutrino Rung Positions)**.** The neutrino mass eigenstates occupy deep negative rungs on the φ-ladder:

State	Integer baseline	Fractional residue	m_ν (eV)
ν_3	$R_3 = -54$	$r_3^{\text{frac}} = -217/4$	≈ 0.056
ν_2	$R_2 = -58$	$r_2^{\text{frac}} = -231/4$	≈ 0.008
ν_1	$R_1 \approx -60$	$r_1^{\text{frac}} = -239/4$	≈ 0.003

The generation spacing in the integer baseline is $\Delta \approx 4$ rungs (half the 8-tick epoch).

Remark 4.2 (Why Deep Negative Rungs?). The neutrino rungs ($R \sim -54$ to -60) are far below the charged lepton sector baseline ($\ell_{\text{lepton}} = 2$). The depth arises because neutrinos, having zero charge, receive no gap-function boost and occupy the lowest-energy region of the φ-ladder. This is the RS explanation for the smallness of neutrino masses: they are not suppressed by a heavy mediator (as in the seesaw), but simply sit at the bottom of the same ladder that accommodates all fermions.

5 Mass Predictions and Oscillation Data

Proposition 5.1 (Squared Mass Differences — Numerical Evaluation)**.** *The predicted squared mass differences are:*

$$\Delta m_{32}^2 \equiv m_3^2 - m_2^2 \approx (0.056)^2 - (0.008)^2 \approx 3.1 \times 10^{-3}\ \text{eV}^2, \tag{4}$$

$$\Delta m_{21}^2 \equiv m_2^2 - m_1^2 \approx (0.008)^2 - (0.003)^2 \approx 5.5 \times 10^{-5}\ \text{eV}^2. \tag{5}$$

Remark 5.2 (Comparison with Experiment).

Parameter	RS prediction	NuFIT 5.2	Agreement
Δm_{21}^2	5.5×10^{-5} eV^2	$(7.53 \pm 0.18) \times 10^{-5}$	$\sim 27\%$ off
$\lvert\Delta m_{32}^2\rvert$	3.1×10^{-3} eV^2	$(2.453 \pm 0.033) \times 10^{-3}$	$\sim 26\%$ off
Ordering	Normal	Favored ($\sim 2.5\sigma$)	Consistent

The RS predictions agree at the correct order of magnitude but show $\sim$ 25–27% discrepancies with the experimental central values. Given the $\sim$ 1–2% experimental precision, these are significant quantitative mismatches that reflect the limitations of the fractional rung assignments. The rung numbers ($-217/4$, $-231/4$, $-239/4$) are determined by matching to observation (not derived from first principles), so these discrepancies indicate that the current rung assignments are approximate. A first-principles derivation of the neutrino rung positions from the RS Hamiltonian is an identified open problem.

5.1 Mass Ratio

Proposition 5.3 (Neutrino Mass Ratio)**.** *The mass ratio between the heaviest and middle neutrino eigenstates:*

$$\frac{m_3}{m_2} = \varphi^{r_3^{\text{frac}} - r_2^{\text{frac}}} = \varphi^{(-217/4)-(-231/4)} = \varphi^{14/4} = \varphi^{3.5} \approx 6.85. \tag{6}$$

Remark 5.4. The observed ratio $m_3/m_2 \approx 0.056/0.008 \approx 7$ is consistent with $\varphi^{3.5} \approx 6.85$. The neutrino generation spacing of ~ 3.5 fractional rungs is roughly half the integer baseline spacing of 4 rungs, reflecting the same "half-epoch" pattern ($4 = 8/2$) seen in other fermion sectors.

6 Why Neutrinos Are Light: No Seesaw Needed

Remark 6.1 (Sub-eV Scale from Deep Rungs). Neutrinos are sub-eV because they occupy deep negative rungs on the φ-ladder ($R \approx -54$ to -60), far below the charged lepton sector ($R \approx 2$ to 19). No seesaw mechanism is required: the mass hierarchy between charged and neutral leptons is encoded in the rung separation.

Note that a naive estimate of $m_e/m_{\nu_3} \sim \varphi^{\Delta r}$ using only the rung difference is complicated by the fact that the charged leptons have a non-zero gap function ($\mathrm{gap}(Z_e) \neq 0$), different sector yardstick Y_s, and renormalization-group corrections. The full mass law (1) with all corrections must be used for quantitative mass ratios.

Remark 6.2 (Contrast with the Seesaw). The standard seesaw mechanism generates small neutrino masses via $m_\nu \sim m_D^2/M_R$, requiring a heavy right-handed neutrino with $M_R \sim 10^{14}$ GeV. In RS, no such particle exists—neutrinos are simply on lower rungs of the same ladder. The seesaw hypothesis (heavy Majorana neutrinos, lepton number violation, leptogenesis) is not needed and not predicted.

7 Dirac Nature from Double-Entry Ledger

Proposition 7.1 (Neutrinos Are Dirac — Structural Argument)**.** *The ledger's double-entry structure requires each fermion to have a distinct antiparticle entry. Majorana neutrinos ($\nu = \bar{\nu}$) are excluded.*

Structural argument. In the recognition ledger, every event is recorded as a debit–credit pair: particle creation at voxel **n** (debit) is balanced by antiparticle creation at the conjugate entry (credit). For a Majorana neutrino, $\nu = \bar{\nu}$ requires the debit and credit entries to represent the same object—creating a self-paired entry rather than a proper debit-credit distinction. This self-pairing is excluded by the double-entry structure of the ledger, which requires distinct debit and credit records for every recognition event.

More formally, the J-cost functional satisfies $J(x) = J(1/x)$ (the ledger symmetry under $x \mapsto 1/x$ corresponds to particle-antiparticle exchange). A Majorana field satisfies $\nu = \bar{\nu}$, which maps to a self-inverse ledger entry ($x = 1/x$, i.e., $x = 1$). The unique solution $x = 1$ is the identity (zero cost, the ground state), not a particle excitation. Therefore Majorana neutrinos with $m_\nu \neq 0$ cannot exist in the RS ledger.

Caveat: This argument relies on the identification of lepton number with a conserved ledger charge. A complete proof from the RS Hamiltonian that lepton number is strictly conserved (and not merely approximately conserved) is an identified open problem. □

Corollary 7.2 (No Neutrinoless Double-Beta Decay)**.** *Since neutrinos are Dirac, the lepton-number-violating process $0\nu\beta\beta$ ($nn \to pp + 2e^-$) is forbidden.*

Remark 7.3. Current experiments (KamLAND-Zen [10], GERDA, LEGEND, nEXO, CUPID) search for $0\nu\beta\beta$ with half-life sensitivity reaching $\sim 10^{26}$ years. RS predicts null results in all such experiments. Any confirmed observation of $0\nu\beta\beta$ would falsify the RS prediction of Dirac neutrinos and the double-entry ledger structure.

8 Normal Mass Ordering

Proposition 8.1 (Normal Hierarchy — Structural Prediction)**.** *RS predicts normal mass ordering:* $m_1 < m_2 < m_3$.

Proof. The fractional rung assignments satisfy $r_1^{\text{frac}} < r_2^{\text{frac}} < r_3^{\text{frac}}$ (i.e., $-239/4 < -231/4 < -217/4$). Since $\varphi > 1$, the mass $m = Y \cdot \varphi^r$ is a strictly increasing function of r. Therefore $m_1 < m_2 < m_3$.

Inverted hierarchy would require $r_3 < r_1$, contradicting the generation-to-dimension assignment where generation 3 always receives the highest rung within each sector. □

Remark 8.2. JUNO [7] and DUNE [8] will determine the mass ordering with $> 3\sigma$ significance within the next decade. Current global fits favor normal ordering at $\sim 2.5\sigma$ [6].

9 Mass Sum and Cosmological Bounds

Proposition 9.1 (Mass Sum Prediction)**.** *The total neutrino mass sum:*

$$\sum_{i=1}^{3} m_{\nu_i} \approx 0.056 + 0.008 + 0.003 = 0.067 \text{ eV}. \tag{7}$$

Remark 9.2. The Planck 2018 + BAO cosmological bound is $\sum m_\nu < 0.12$ eV at 95% CL [5]. The RS prediction of 0.067 eV is well within this bound. Next-generation surveys (DESI, Euclid, CMB-S4) are expected to reach sensitivity of $\sum m_\nu \sim 0.02$ eV, which will definitively test the RS prediction.

10 PMNS Mixing from Rung Separations

Remark 10.1 (Mixing from Path Costs — Structural Estimate). The PMNS mixing matrix elements are expected to arise from the J-cost differences between neutrino rung positions. The structural estimate is:

$$U_{ij} \sim \exp(-\Delta r_{ij} \cdot J_{\text{bit}}), \tag{8}$$

where Δr_{ij} is the rung separation between flavor state i and mass eigenstate j, and $J_{\text{bit}} = J(\varphi) = \varphi - 3/2 \approx 0.118$ is the minimal recognition cost. The large mixing angles observed in the PMNS matrix (compared to the small CKM angles) are explained qualitatively by the relatively small rung separations in the neutrino sector ($\Delta \approx 4$ rungs) compared to the quark sector: smaller rung separations mean larger off-diagonal mixing. This estimate is *structural* and not a precise quantitative derivation; a full computation of all three PMNS angles and the CP phase from the RS rung structure is presented in a companion paper.

11 Predictions and Falsifiers

1. **Normal ordering**: $m_1 < m_2 < m_3$. Testable by JUNO and DUNE within the next decade.

2. **Dirac neutrinos**: No $0\nu\beta\beta$ decay. Testable by LEGEND, nEXO, and CUPID to half-lives $\sim 10^{28}$ years.

3. **Mass sum**: $\sum m_\nu \approx 0.067$ eV. Testable by Euclid, DESI, CMB-S4.

4. **Heaviest mass**: $m_3 \approx 0.056$ eV. KATRIN [9] will constrain the effective electron-neutrino mass to ~ 0.2 eV, with Project 8 targeting ~ 0.04 eV.

5. **No sterile neutrinos**: $N_\nu = 3$ exactly. No light sterile species mixing with active flavors.

6. **No seesaw particles**: No heavy right-handed neutrinos at any mass scale.

12 Conclusion

Neutrino masses in RS arise from the same φ-ladder mechanism as all fermion masses. The sub-eV scale is explained by deep negative rung assignments ($R \approx -54$ to -60), not by a seesaw mechanism. The key structural features are:

1. Neutrinos have $Z = 0$, so $\mathrm{gap}(0) = 0$: the simplest case of the mass law.

2. The generation spacing is $\Delta \approx 4$ rungs (half the 8-tick epoch).

3. Dirac nature is forced by double-entry ledger conservation.

4. Normal hierarchy is forced by rung monotonicity.

5. Mass sum $\sum m_\nu \approx 0.067$ eV is below current cosmological bounds.

The structural predictions—Dirac nature, normal ordering, no $0\nu\beta\beta$—follow from the ledger and φ-ladder with zero free parameters. The quantitative mass values depend on the rung assignments, which are currently observationally motivated; first-principles derivation of the neutrino rung positions is an open problem.

References

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